

VISUALIZING THE INVISIBLE HAND OF MARKETS: SIMULATING COMPLEX DYNAMIC ECONOMIC INTERACTIONS

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SUMMARY

In complex systems, many different parts interact in nonobvious ways. Traditional research focuses on a few or a single aspect of the problem so as to analyse it with the tools available. To get a better insight of phenomena that emerge from complex interactions, we need instruments that can analyse simultaneously complex interactions between many parts. Here, a simulator modelling different types of economies is used to visualize complex quantitative aspects that affect economic dynamics. The main conclusions are: (1) relatively simple economic settings produce complex nonlinear dynamics and, therefore, linear regressions are often unsuitable to capture complex economic dynamics; (2) flexible pricing of goods by individual agents according to their microenvironment increases the health and wealth of the society, but asymmetries in price sensitivity between buyers and sellers increase price inflation; (3) prices for goods conferring risky long-term benefits are not tracked efficiently by simple market forces; (4) division of labour creates synergies that improve enormously the health and wealth of the society by increasing the efficiency of economic activity; (5) stochastic modelling improves our understanding of real economies, and didactic games based on them might help policy-makers and nonspecialists in grasping the complex dynamics underlying even simple economic settings. Copyright © 2015 John Wiley & Sons, Ltd.

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1. INTRODUCTION

Adam Smith (1977[1776]) and Friedrich Hayek (1961), among many others, assigned extraordinary importance to hidden properties of markets. They assumed the existence of poorly understood complex market mechanisms upon which modern economies are built. These mechanisms and the phenomena they produce are not easily grasped analytically because of their extraordinary complexity. Mathematics, the language we use to ask questions to nature, has expanded the limits of economic analytical power since Hayek wrote about the ‘Economic Calculus’ and Smith about ‘The Invisible Hand’. We now are capable of viewing economies with more powerful tools that allow for a more rigorous analysis of complex features formerly regarded as off limits to rational reductionist methods by economists. These tools allow one to produce multidimensional radiographies of an economy. They include ‘cellular automata’ (Axelrod, 1984), ‘active walks’ (Lam, 2005, 2006) and computer simulations. Specifically, agent-based modelling (ABM) allows one to simulate very complex phenomena in economics (e.g. Prietula & Carley, 1994; Kochugovindan & Vriend, 1998; Magliocca, Brown, & Ellis, 2014). ABM simulations can easily be made so complex as to be beyond our understanding, becoming unpredictable. Thus, in order to remain useful, we have to limit the complexity of ABM so as not to trespass this edge of chaos. The mathematically relevant challenge is to find the simplest representation of reality that is able to capture the problem to be studied.

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Sociodynamica is an agent-based simulation model, purposely built for analysing economic microdynamics. It is based on classical monetary economic principles (Gurley & Shaw, 1960; Wray, 1996) and builds virtual societies of agents that collect and trade two types of heterogeneously distributed resources. It can be made as complex as one would wish, but in order to comply with the maximum entropy principle (Jaynes, 1957) we want to keep the problem simple and advance our knowledge in small steps. The aim here is to build the simplest mathematical construct that still reflects aspects of the economic dynamics that are fundamental to modern complex economies. Specifically, we want to reproduce phenomena, such as the emergence of the ‘Invisible Hand’ coined by Adam Smith. Sociodynamica is specifically aimed to visualize the process of emergence of new macro phenomena from the microscopic social interaction of agents and has been in continuous development, starting from Biodynamica, for the last 20 years. A large number of alternative agent-based simulation models for economic problems are available—see Tesfatsion (2014). For example, ‘Sugar-scape’, an agent-based model developed independently by Axelrod (1984), and the model developed by Axtell (2005) are very similar to Sociodynamica. Eventually, repeating the simulations presented here with other agent-based models, and devising empirical studies based on these simulations, should add confidence to the results presented and advance our understanding of complex markets.

2. METHOD

Sociodynamica is a freely available agent-based simulation model written in Visual Basic in which diverse agents roam a virtual landscape looking for resources. Agents farm and mine for foods and minerals and trade their surplus according to different economic settings. Sociodynamica assumes that utility functions in economics are equivalent to fitness functions in biology (Kenrick *et al.*, 2009), simulating the survival of agents in situations with or without heredity and with or without movement of agents. Sociodynamica has been used previously to study the effect of altruism and altruistic punishment on aggregate wealth accumulation in artificial societies (Jaffé, 2002a, 2004a, 2008; Jaffé & Zaballa, 2009, 2010) and in studying the dynamics of complex markets (Jaffé, 2002b, 2004b).

The present model simulated a virtual society of agents that farmed and mined food and minerals respectively and traded their surplus according to different economic settings. The agents inhabited a continuous, flat two-dimensional toroidal world (see Figure 2) that was supplied with patches of agricultural land (‘sugar’ or ‘food’) and separate nonoverlapping patches of mines (‘spices’ or minerals). Immobile agents were randomly dispersed over this fine-grained virtual landscape. Each time step, agents that happened to be located over one of the resources acquired a unit of the corresponding resource, accumulating wealth, either as sugar or food (G_1) and/or as spices or minerals (G_2). Agents spend a fixed amount of each resource in order to survive, consuming each of them at a basal constant rate (default value was 0.1 units of the corresponding resource). Both resources were consumed and metabolized similarly, but food was three times more abundant than minerals (the size of the patch for minerals was normally set to 100×100 pixels and for food 300×300 pixels). Each patch remained in the same place during each simulation run and the resources inside them were replenished continuously. Agents perished when they exhausted any of the two resources. The population of agents was maintained constant by introducing after each time step new agents with randomly assigned initial parameters. Initial parameters were the type of agent, the random spatial position and the initial amount of money used to start trading resources (the default initial value was set to 10 units of money). The amount of money each agent possessed varied according to its trade balances. Agents gained money when selling food and/or minerals and spent money when buying them.

Agents traded the resources they possessed with other agents. They sold the resource they possessed in more abundance and bought the one they were short of. In simulations without 'division of labor', the trade could take place between any agent in a population of omnipotent agents. In simulations with division of labor, agents traded according to their specialization. Here, agents were subdivided into three categories: Farmers, who specialized in collecting only Food (Resource 1); Miners, who extracted only Minerals (Resource 2); and Traders, who specialized in trading minerals for food when encountering a Farmer and trading food for minerals when encountering a Miner. Farmers traded only with Miners and Traders; Miners traded only with Farmers and Traders. Traders could interchange resources with all types of agents. Trades were allowed only between agents spaced at a distance not larger than the 'contact horizon' of the trading agent. Each time step, buyers searched for potential sellers of the required good by contacting randomly up to 10 agents in the area defined by this contact horizon. If the agent found another agent with the wanted good willing to sell at or below the price defined by the buyer, a trade was executed using the price of the seller. Trades were limited to the amount of money available to the buyer and the amount of goods possessed by the seller, unless credit was simulated. Variation of the contact horizon allowed the simulation of different levels of globalization or integration of economic agents. Sellers not able to sell their product during one such trading session decreased their price for the resource; and buyers unable to get what they wanted at the preferred price increased the price they were willing to pay in the next round of trades.

Sociodynamica can simulate biological and/or cultural evolution. Both have similar dynamics (Jaffe & Cipriani, 2007). Here, we simulated cultural evolution in economic markets. This evolutionary process weeded agents with an unfit combination of parameters, revealing the optimal set for each economic scenario. The time that immobile agents survived was reflected in the average age of agents, or 'Age', and was used as an estimate of the 'Health' of the economy. The 'Wealth' of the economy was assessed by the total resources (Food and Minerals) accumulated by surviving agents.

2.1. Types of Resources (Res)

Two different types of interactions between agents and resources were modelled. To make results comparable to popular alternatives to Sociodynamica, we simulated the setting of 'Sugar and Spices' (SS) by Axelrod (1984). In simulations of SS, both resources were consumed and metabolized similarly. When simulating Food and Security (FS), Security was provided by Resource 2. G_2 , the wealth in Resource 2, was inversely related to the probability of sudden death for each agent. That is, Resource 2 improved the odds of surviving external 'catastrophes' that killed agents at random from time to time. Large amounts of G_2 protected the agents against these catastrophes by reducing the probability of being affected by them so that agents survived if $G_2 - \text{Rnd}(0-1) * D > 0$.

This algorithm achieved that the greater the wealth of accumulated minerals of the agent, the lower their probability of being struck by a catastrophe, at any level of danger D .

Money served to trade resources. Each agent was endowed with an initial amount of money (10 units, unless stated otherwise), which was used to pay for sales and to provide credit.

2.2. Simulations

Different aspects of an economy were modelled, including:

1. Credit for agents lacking money (TE1).
2. Biased influence of prices biased towards sellers (TE2).

3. Biased influence of prices biased towards buyers (TE3).
4. Different relative importance between the two resources (abundance regulated through patch size or consumption regulated by metabolism).
5. Different geographical (spatial) distribution of resources (topology of the distribution).
6. Different balances between income and use of resources (balancing productivity of resource with its consumption).
7. Different types of resources—Sugar and Spices where the two resources behaved similarly, or Food and Security where the effect of one resource on fitness was constant and the other (security) was random and variable.
8. Prices an agent was willing to pay for a given resource increased after each failed attempt to buy and decreased after each failed attempt to sell the respective resource. In some simulations specifically mentioned below, agents in addition increased the price to sell after a successful transaction or decreased the price to buy after a successful shopping.
9. Combinations of the above.

Economic equilibrium of a given modelled system was assessed using the following variables.

Balance between income I of resources r and their consumption C . For survival, agents were required to conform to

$$Ir > Cr$$

Income can be either by direct gathering G or by trade T :

$$Ir = Gr + Tr$$

Here, each agent has to balance two resources in order to survive. I simulated a utility function U so that U had to remain positive for the agent's survival and

$$Ur = Gr + Tr - Cr$$

The amount of traded resources can increase or decrease, according to the balance of resourced bought B and sold S :

$$Tr = Br - Sr$$

The amount of resources bought and sold depends on the availability of money M and the price P paid for the resource by the agent a :

$$Br = Ma/Pr_a \text{ and } Sr = Ma/Pr_a$$

The amount of money of agent a depends on the amount spend buying (Mbr) and the amount gained selling (Msr) resources

$$M_a = M_{a0} + Msr - Mbr$$

where M_{a0} was the initial amount of money supplied to each new agent j .

The computer simulations consisted in numerically calculating Ur_a every time step for the population of agents a , so that Wr , the total accumulated wealth for each resource, is

$$Wr = \sum_a Ur$$

Agents with $U_1 \leq 0$ or $U_2 \leq 0$ were eliminated and substituted by new ones with default properties, as an analogy of broken companies that are replaced by new start-ups.

Results are presented with only four variables that reflect the state of the simulated economies:

1. **TAR**—total amount of each resource accumulated by the population; $TAR = W_1 + W_2$.
2. **Age**—average age of agents measured in Time steps.
3. **FPr**—average price for Resource 1, called Food, equivalent to either Sugar or Food.
4. **MPr**—average price of Resource 2, called Mineral, equivalent to either Spices or Security.

The variables or features of the simulations explored here are summarized in Table 1.

The detailed program in Visual Basic is available in the help feature of the program. Simulations can be run with parameters chosen at will by downloading Sociodynamica at <http://atta.labb.usb.ve/Klaus/Programas.htm>. Here, I present the results of only 16 types of simulations, each type run at least 200 times for 200 time steps, as a representative sample of the effects of the various parameters studied. Default values, unless explicitly stated otherwise, were:

transaction horizon	200
initial prices for both resources	3 units
reserve units not traded	1
patch size for Resource 1	300
patch size for Resource 2	100
resources metabolized per time step	0.1
resources collected per time step if placed over the resource	2
number of agents	500
maximum number of trades per time step	10

More details are given in the Appendix. Videos of simulations are available at <http://atta.labb.usb.ve/Klaus/EC/ECVideos.html>.

3. RESULTS

The simulator allowed the design of experiments where several different variables interacted with each other simultaneously, so as to study complex interactions between several factors affecting the economy

Table I. Features explored in the simulations

SS	Resources behaved as in Sugar and Spices
FS	Resources were Food and Security, and were Minerals protected against random culling
Types of agents	<i>Omnipotent</i> : all agents collected and traded both types of resources <i>Division of labour</i> : agents were Farmers or Miners, specialized in collecting and selling food or minerals, or Traders specialized in trading both resources
FMar	Price adjustment in monetary units: 0 is equivalent to fixed prices and 0.5 when prices were flexible
FPro	Flexible productivity so that increased prices increased production of resource
TE	Type of economic feature simulated TE0: control simulations with default parameters TE1: credit TE2: agents finishing a successful sale, increase price by one unit TE3: successful buyers decrease their future asking price by one unit TE4: TE2 + TE3
Danger <i>D</i>	Each time step, 10% of the population was killed, unless another value is given. Relative wealth in minerals protected agents. The algorithm used to trigger culling was $100 * Rnd/U2/W2 < 1000 * Rnd/D$. <i>D</i> represented the perceptual value assigned for culling
Credit	When agents buying from a trader or from an omnipotent agent had less money than required to finish the transaction, they asked for a loan. The trader gave half of the amount of the resource requested for trade to the buyer and the buyer increased its depth according to the price of the resource given by the seller. After this trade, each time the buyer met a trader and if it had the money, it paid two monetary units to this trader, which could be different to the former one, until the depth was cancelled.

at the same time. The results of over 200,000 simulations exploring the effect of dozens of interrelated variables cannot be presented here in detail. For a fuller exploration of the simulations, the reader is advised to download the simulation program and design experiments on their own. Only a selected summary of lessons learned from Sociodynamica's simulations is present. These can be grouped into results that confirm known economic dynamics, validating the model (results 1–3); results that are relevant for understanding economic microdynamics (results 4 and 5); and novel serendipitous or unexpected findings (results 6 and 7).

3.1. Result 1: Wealth Distribution among Agents Mirrors That of Empirical Data

The wealth distribution among agents is shown to be a negative exponential function with fat tails (Figure 1). This form of the curve is indistinguishable from those found empirically by economists and physicists—for example, see Levy and Solomon (1997); Dragulescu and Yakovenko (2000); Gonzalez-Estevez, Cosenza, Lopez-Ruiz, and Sanchez (2008) and Lopez-Ruiz, Lopez, and Calbet (2012). The exact form of the curve varied somewhat, showing slight differences when obtained in simulations with differed parameters. But all distributions obtained showed the same basic power distribution with long tails. This distribution differs from a normal distribution, which is assumed by default by most classical economists when analysing economic data. Averages and parametric statistics are not very meaningful in analysing data with these distributions. Nonparametric statistics are more adequate for handling this kind of data.

3.2. Result 2: Heterogeneous Distribution of Resources Affects Price Dynamics

Prices, when allowed to be adjusted by individual agents, were heterogeneously distributed among the population of agents, and sellers of a resource had higher price expectation than buyers (Figure 2). That is, sellers of sugar were more likely to expect sugar to be dearer than spices, and sellers of spices were

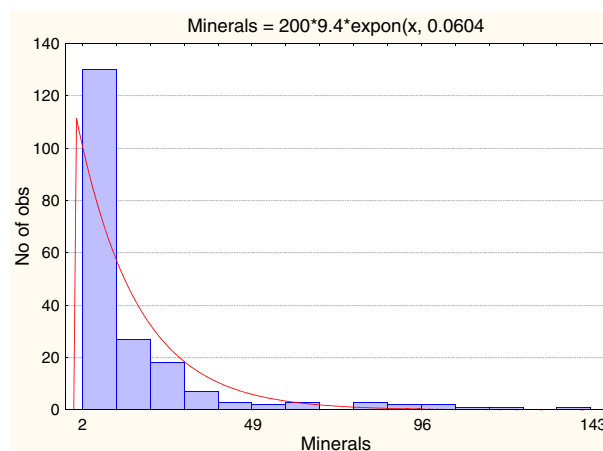


Figure 1. Example of frequency distribution among 2000 agents in respect to different amounts of goods possessed by each agent in a selected simulation

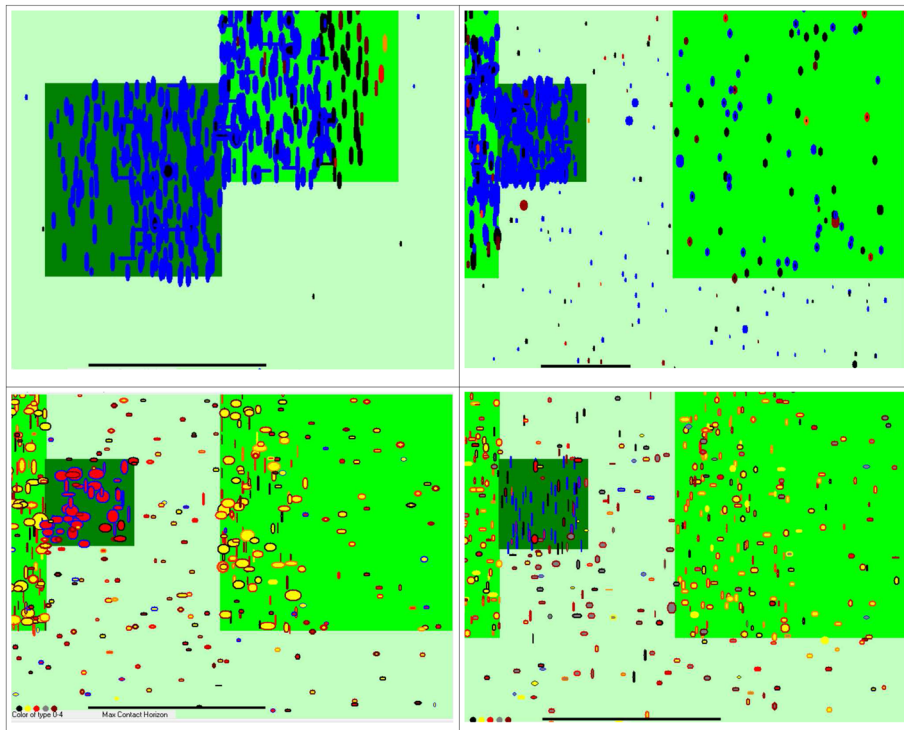


Figure 2. Examples of total resource landscape and amplification of a section showing agents. First row: examples of simulations EC01, EC02; second row: EC09, EC15. Colors in print are represented in different shades of gray

- Bright green field is covered with ‘Food’; darker green field is covered with ‘Minerals’; the lightest green is devoid of resources.
- The size of the bubble is proportional to the wealth of the agent, so that the thickness is determined by $(Ur_1 + Ur_2)$ and the height by the amount of money the agent possesses.
- The thickness of the border is proportional to the perceived cost of living, calculated as $FPr + MPr$.
- The colour of the border is more reddish or even yellow the higher the ratio FPr/MPr . That is, agents with red and yellow pay more for minerals, whereas those with blue or black pay less for minerals compared with what they are willing to pay for food. For simulations EC001 and EC002 using omnipotent agents with no division of labour, this same colour was applied to the whole body. For the rest of the simulations, the colour of the body was proportional the colour of the body of sphere described the type of agent so that yellow were Farmers, red were Miners and grey were Traders.

more likely to expect spices to be dearer than sugar. This effect was due to the fact that prices increased with scarcity. Prices for the two goods were similar (simulations achieved for both prices of 1.2 units after 200 time steps; a t -test revealed no difference between both prices at $p < 0.1$) when land with spices had the same area as land covered with sugar. When land covered with sugar had three times the area as land covered with spices, simulations showed that prices for the scarce resource, spices, were higher (1.7 units after 200 time steps) compared with prices of the more abundant one, sugar (1.2 units). The difference was statistically highly significant ($p < 0.0001$ using a t -test comparing both prices).

3.3. Result 3: Spot Prices Are Much More Volatile Than Average Prices

The model allowed one to track simultaneously the spot prices and the average acceptable prices of all agents. This is normally not possible in real economies. The spot price was calculated from the last two transactions after every time step, whereas averages were calculated by taking into account all transactions during a time step. Average prices was an estimate of the prices agents were willing to pay in a commercial transaction. Spot prices and average prices differed significantly (Figure 3). The specific distribution of price differences between food and minerals depended on the economic topology of the system, but spot prices were always more volatile. The long-term average for spot prices was similar to the average prices for each resource in the population. However, when selecting randomly a given spot price, chances were that spot prices differed greatly from the average prices.

3.4. Result 4: Elements Affecting the Wealth and Health of an Economy

Table 2 presents the effect of the different features simulated on the health and wealth of the virtual economy. Health was assessed using 'Age', which reflects the quality of life for the average agent, measured through the average age of the agents. Age was correlated to a measure of wealth of the economy called 'TAR', which represents the total amount of resources (food and minerals) accumulated. The most important features revealed by the results in Table 2 were as follows.

Result 4.1: Synergy Is Created by Division of Labour

When we compare the results of simulation where agents divided their labour with those where agents were omnipotent, we observe that division of labour increased price differentials, longevity and wealth. Specifically, there was a very substantial difference in TAR and Age. The division of labour created synergies in the economic dynamic, fomenting useful trades (i.e. between collectors of food with collectors of minerals) and avoiding useless trades (i.e. between collectors of the same resource type). The difference was higher among simulations using SS settings than among those using FS settings.

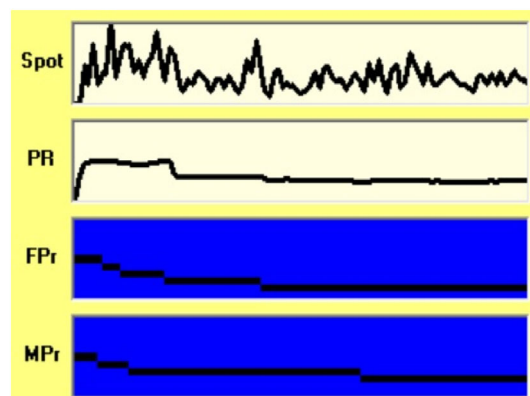


Figure 3. Spot and PR are the ratio of Food Price (FPr)/Mineral Price (MPr) for the last two transactions (Spot) or for the average of the whole population of agents

Table II. Wealth (TAR in thousands of units) and health (Age in time steps of the average agent) in different economies with 500 agents after 200 time steps^{a,b}

			TAR					Age					
Res	FMar	FPro	TE0	TE1	TE2	TE3	TE4	TE0	TE1	TE2	TE3	TE4	Sim
<i>Omnipotent agent</i>													
SS	n	n	3	8				2	10				EC04
SS	y	n	4	8	3	6	4	4	9	2	6	3	EC05
SS	y	y	5	4	4	3	5	11	8	4	7	9	EC13
FS	n	n	5	4				7	5				EC06
FS	y	n	5	4	5	5	5	7	5	6	6	7	EC07
FS	y	y	3	3	6	3	4	7	5	6	5	7	EC12
<i>Division of labour</i>													
SS	n	n	58	63				73	78				EC08
SS	y	n	49	44	51	48	51	78	93	71	76	73	EC09
SS	y	y	11	9	44	3	25	89	96	70	64	74	EC15
FS	n	n	8	9				17	17				EC10
FS	y	n	4	4	6	3	4	15	17	16	14	15	EC11
FS	y	y	2	2	7	1	3	14	15	18	14	13	EC14

^aRes: the type of resources simulated (SS for sugar and spices and FS for food and security); FMar indicates if prices could vary; FPro indicates if productivity was flexibly regulated by prices. Omnipotent agents simulate no division of labour. Sim refers to the specific simulation.

^bTE0: control; TE1: credit; TE2: agents finishing a successful sale, increase price by one unit; TE3: successful buyers decrease their future asking price by one unit; TE4 = TE2 + TE3; TE5 = TE4 + TE1. FMar: y then prices changed 0.5 units each time a trade failed. Simulations of TE2 to TE4 are not possible when prices are fixed (FMar = n).

Result 4.2: Types of Resources Simulated

The types of resources simulated affected the outcome of the simulation greatly. Simulating sugar and spice (SS) produced different outcomes compared with simulations with food and security (FS). Wealth levels achieved by agents and their longevity were much larger with SS than with FS. However, the decrease in longevity caused by FS was worse without division of labour. Simulations with omnipotent agents (i.e. without division of labour) produced much lower levels of wealth and longevity.

Result 4.3: Credit

Simulating credit increased the accessibility of resources significantly when omnipotent agents were simulated (EC04, EC05). In contrast, the effect of credit was very small when division of labour was modelled in the simulation (EC08, EC09). That is, the positive effect of credit was less evidenced when the positive effect of synergies triggered by division of labour was present. When prices were flexible, the positive effect of credit was reduced or eliminated (e.g. EC5 versus EC13), reflecting the fact that improved commercial transactions are better achieved by flexible pricing than by increasing money supply through indiscriminate credit as modelled here.

Result 4.4: Correlations between Wealth and Health

In general, wealth and health reflected similar aspects of the economy. The correlation between both variables in the data shown in Table 2 is high ($r=0.82$), though both measures were not identical. In some cases 'Age' and 'TAR' were not correlated. An interesting effect can be observed in economies with division of labour, where a fuller economic development was achieved: focusing on TE0, we observe that flexible productivity decreased TAR but increased Age (EC09 versus EC15). In contrast, flexible prices (FMar = y) increased both health and wealth of the societies (compare EC10 and EC15).

3.5. Result 5: Elements Affecting Prices

The dynamics of prices differed from that of TAR and Age. In general, prices behaved as expected (Table 3). Some interesting features about prices revealed by these simulations were as follows.

Heterogeneous Prices Reflect the Efficiency of an Economy

Division of labour increases the price difference between Food and Mineral. Mineral prices were always higher than Food prices. This reflects the fact that the smooth working of an economic system helps produce better pricing, which in turn favours the economy. This explains why prices of both resources increased with FS simulations compared with SS simulations, as with FS type of resources the economic dynamic was more chaotic.

Credit Did Not Affect Prices

Although credit affected the health and wealth of society (Table 2), it did not affect prices (Table 3). Credit increased the pool of total resources accumulated by agents and their average age, but had little effect on prices (compare values under TE0 and TE1).

Psychology of Traders Affect Prices

The disposition to lower the price at which an agent was willing to buy in the next round of trades had a larger effect on the dynamics than price increases implemented by successful seller when buyers outnumbered sellers. In Table 3, prices in the column for TE2 were always higher than with TE3, independently of the balance between supply and demand. Asymmetries in price sensitivity between buyers and sellers will produce above-average price inflation. The results showed that stingy sellers increased prices and less generous buyers decreased them.

3.6. Result 6: The Degree of Globalization (i.e. the Contact Horizon of Agents Affects Prices Depending on the Topology of the Resource Distribution)

In this simple model, the topology of the resource distribution had an effect on prices (Figure 4). The results showed that globalization reduced prices if the increased contact horizon did not introduce additional topological barriers. Price levels for the scarce resource (minerals) were higher when the commercial horizon of the agent was smaller when barriers to trade were absent (i.e. agents could easily bridge the gap between patches with food and patches with minerals). Overall economic wealth and

Table III. Prices for Food (FPr) and Minerals (MPr) in simulations in Table 1

Res	DL ^a	FPro	FPr					MPr					Sim
			TE0	TE1	TE2	TE3	TE4	TE0	TE1	TE2	TE3	TE4	
SS	n	n	0.9	0.9	1.7	0.6	1.0	1.1	1.1	2.0	0.7	1.2	EC05
SS	n	y	1.0	0.9	1.6	0.5	1.0	2.2	1.0	3.4	0.5	2.0	EC13
FS	n	n	1.0	0.9	2.9	0.6	1.2	1.6	1.2	2.8	0.6	1.1	EC07
FS	n	y	0.9	0.8	2.3	0.5	1.2	1.4	1.1	2.2	0.6	1.2	EC12
SS	y	n	1.0	0.9	1.8	0.5	0.9	3.4	3.0	2.9	0.6	1.0	EC09
SS	y	y	0.8	0.8	1.8	0.5	0.9	3.2	3.1	2.8	0.5	1.0	EC15
FS	y	n	0.9	0.9	2.0	0.5	1.0	4.3	4.8	3.1	0.6	1.0	EC11
FS	y	y	0.8	0.8	2.5	0.5	1.1	3.7	3.9	3.9	0.6	1.0	EC14

^aDivision of labour.

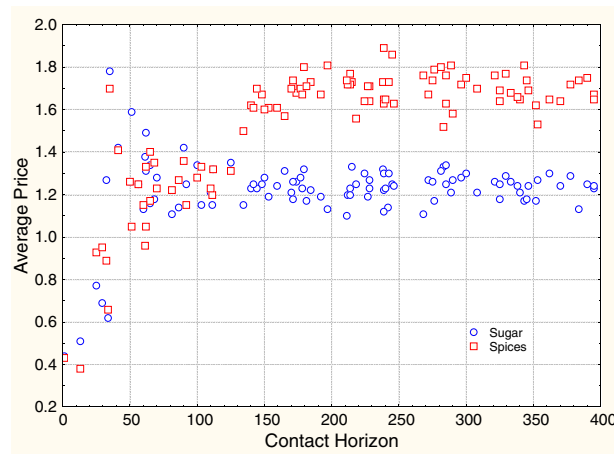


Figure 4. Examples of the dynamics of price for goods (circles: Sugar or Food; squares Spices or Minerals) reached after 200 time steps with different maximum contact horizons for agents in simulations of sugar and spices

longevity of agents increased with increased contact horizon (Figure 5). The distribution of data in Figures 4 and 5 shows that linear regressions do not capture the important elements of this dynamics.

Increasing the contact horizon of agents increased the odds of an agent to complete a successful trade. Thus, synchrony in economic transactions increased with increased contact radius. This fact explains the results obtained.

3.7. Result 7: The ‘Invisible Hand’ Proposed by Adam Smith Can Be Made Visible, Allowing the ‘Economic Calculus’ Supposed to be Impossible by Friedrich Hayek

Among the various factors studied, two were most relevant regarding the objective of visualizing emergent phenomena in the market. These factors were flexible pricing and division of labour. Each of these factors revealed nonintuitive features in the economic dynamics.

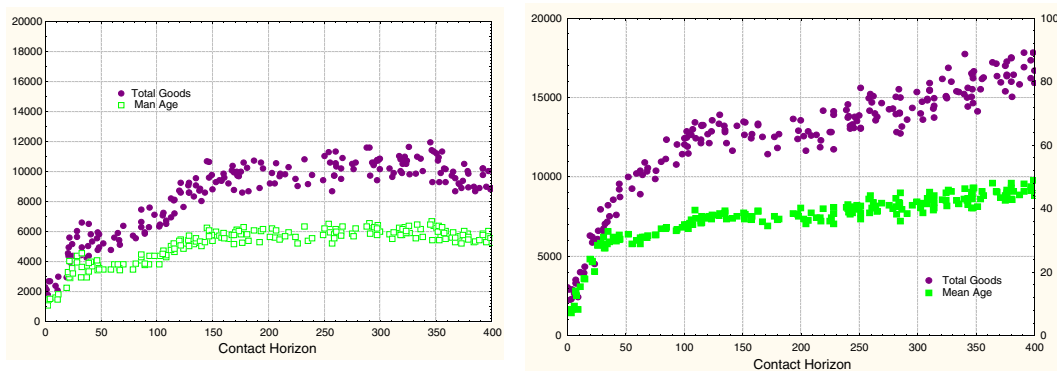


Figure 5. Total production of goods (Food+Minerals) and mean age achieved by economic agents always increased with increasing maximum contact horizons (EC7)

Price Flexibility Protects the Individual, and More So When Credit is Available

Table 4 focuses on the effect of free markets (agents could adjust prices: Free Mar), compared with markets with fixed prices for both resources (no Free Mar). The results are unexpected. Free pricing in markets increased the health of a population (Age) but decreased its overall wealth (TAR). This is one of the rare situations when TAR and Age were not correlated. The presence of credit (TE1) increased this contrasting difference. That is, price flexibility favoured long-term survival of individuals, reflected in an increased average longevity, but reduced the overall aggregate wealth. This effect can be explained by the fact that, with flexible prices, individuals can adjust their accessibility to each of the two resources, maximizing their wealth depending on the environment in which they were located: Miners paid more for Food and Farmers more for Minerals, maximizing their utility. With fixed prices, optimal trading habits could not be adapted to the microenvironment. But flexible prices made some resources inaccessible to some agents, reducing the total amount of accumulated resources.

Division of Labour Creates Economic Synergies That Are Difficult to Predict

It could be argued that omnipotent agents are better prepared to be successful in any given economic environment as they can adapt to a wider range of circumstances, thanks to the fact that they can act as miners, farmers and traders. This, however, did not turn out to be the case. The simulation of division of labour, compared with simulations with omnipotent agents, increased the health and wealth of the population enormously (Table 5). This effect was the strongest among all the contrasting combinations of possibilities explored. Its origin derives from the fact that, in simulations with division of labour, agents avoid selling resources they do not collect. That is, Farmers sold only food and Miners sold only minerals. Traders sold whatever resource they had in more abundance. The presence of traders helped populate spaces not occupied by any of the resources, as was also the case when omnipotent agents were simulated. This, though, seemed to add little to the overall wealth of the populations.

This result shows that, even in this simple example, optimal economic strategies differ for different agents. That is, Farmers do best if they only sell to Miners, and Miners do best if they only sell to

Table IV. Simulations with fixed prices (EC08) compared with simulations with freely changing prices (EC09). Both simulations had division of labour with three agents and SS type of resources

FMar	TAR		Age		Sim
	TE0	TE1	TE0	TE1	
n	58	63	73	78	EC08
y	49	44	78	93	EC09

Table V. The effect of division of labour on the wealth and health of populations exploiting SS resources

	DL ^a	TE0
<i>Age</i>		
EC05	n	4
EC09	y	78
<i>TAR</i>		
EC05	n	4
EC09	y	49

^aDivision of labour.

Farmers. Intermediate strategies failed to reap maximum wealth. This divergence of optimal strategies explains the success of division of labour, as in areas with minerals only Miners survived, and in areas with Food only Farmers survived after a few time steps, producing a division of labour adjusted to the heterogeneities of the heterogeneous distribution of resources.

4. DISCUSSION

The simulations presented here show that the simulator is a powerful analytical tool, producing a rich collection of outcomes that reflect meaningful dynamic aspects of economies. So rich in fact that here I could only present a few of the relevant results. I hope that these are convincing enough to show the potential of such a simulator to motivate game developers to work on them. Many aspects need further refinement of course. For example, the way I simulated credit does not reflect how credit normally works in an economy. However, the rudimentary credit simulated worked in synergy with several other aspects of the virtual economy.

A general insight from this exercise is that, even in this simplest of meaningful economic model, important nonlinear instabilities were present, producing in many cases a nonintuitive dynamic outcome. This contrasts with the widespread use of multiple linear regressions in classical economic analysis. If strong nonlinear dynamics can be evidenced in very simple economic scenarios, then nonlinear dynamics should be expected in real complex economic scenarios. Many examples of analysis of empirical data reveal this fact—for example, see Jaffe, Rios, and Florez (2013).

A most remarkable aspect of Sociodynamica is that it captures the mysterious synergies achieved by the division of labour, as recognized long ago by Adam Smith. Modelling quantitatively the synergy that emerged when simulating division of labour could be very powerful in deepening our understanding of real economies. Few empirical research studies focus on these aspects, as interest among classical economists and business managers seems to lie more on organization and management than on spontaneous emergence of order. Quantitative work on the effect of division of labour on the energy consumption of ants revealed its large potential economic impact (Jaffe & Heblin-Beraldo, 1993). The fact that the insight by Adam Smith, that the 'Invisible Hand' manages markets based on this division of labour, could be confirmed numerically with Sociodynamica is important. Division of labour unleashes synergies from the interactions of agents that can otherwise not emerge. Focusing on the nature and dynamics of these synergies should be an important task for future economists. The present consensus is that strong institutions and rules favour the unleashing of synergies of a free market (World Bank, 2010). Empirical evidence, however, shows that the relative importance assigned to basic scientific knowledge is a much better predictor of present economic strength and of future economic growth than any of the indexes estimating more classical economic variables like strength of institutions, industrial development, education and others (Jaffe *et al.*, 2013). That is, precision and depth in knowledge seems to as important or even more so than social and political structures in building economic synergies.

The emergence of collective phenomena has its own dynamics, as was illustrated here with the effect that the asymmetry of price sensitivity between buyers and sellers has on the economic dynamics. Some examples from real life can illustrate this point. In most countries, drinkable water is more expensive than cheap wine or fizzy drinks, despite the fact that water is the more abundant resource. Results from Sociodynamica suggest that this phenomenon might be explained by putative asymmetries in price sensitivities between consumers' need to quench their thirst and sellers managing their stock of water supply rationally. These asymmetries might explain the above-average price inflation in health

care where the urgency of buyers contrasts with the secure position of sellers. The model shows the probable outcome of a combination of greedy health providers and frightened patients.

A relevant insight gained here relates potentially to insurance schemes. Sociodynamica showed that markets for risky utilities, such as provided by insurance, are not efficiently tracked by simple market forces. This might explain the known differences in costs and efficiency of health insurance systems based on market forces and those based on central planning. Novel incentives to make insurance markets more rational might be tested and perfected with simulations before implementation.

These results just confirm Hayek's hunch that 'the curious task of economics is to demonstrate to men how little they really know about what they imagine they can design' (Hayek, 1991). Even very simple abstractions of economies produce emergent phenomena and irreversible dynamics that are more appropriately understood with scientific tools familiar to ecologists than to physicists. This would suggest that our future understanding of real economies will profit more from consilient bridges with the biological sciences, focusing on diversity and stability rather than econophysics or mathematics with its focus on precise quantitative predictions. This cocktail of approaches can be achieved with agent-based simulations as shown here. Our results show an example of the counterintuitive results of complex economic interactions. The differential effect of free markets on the average age of agents, and on the accumulated wealth of the society, mirrors the polemic between economists focusing on the collective benefits of policies (i.e. socialists, for example) and those focusing on individual freedoms (liberal economist). Sociodynamica seems to be able to provide evidence of how both factors work simultaneously in a given society.

Eventually, it might be possible with more sophisticated models to reflect real economic dynamics in a given system. Climate modelling and weather forecasting have taught us that the task of understanding reality can be tamed, partially with complex simulation modes. Even if for specific settings and moments, Hayek's Economic Calculus might have a solution, the main lesson for economists is that economies are more like the weather: complex systems not easy to predict. Thus, interventions into the economic dynamic have to be treated as careful experiments that have to be monitored and reversed if the outcome turns out to contradict expectations. Hopefully, letting central planners and interventionist-prone politicians play games based on these simulations will help them understand the nature of complex dynamics in economics.

APPENDIX A: Sociodynamica creates a virtual society where agents exploit and compete for resources and share resource 1 among them, according to the settings defined by the *internal parameters* and the *external parameters*. The agents may acquire renewable and nonrenewable resources through work; they may accumulate those resources and commercialize them. At the same time, agents may acquire resources through commerce.

Global Parameters

POP: Number of agents (no)

TAR: Aggregate total wealth accumulated by all agents

Simulation logic

Each time step:

Do simulation loop

Matrix:

Eliminate variable types previously defined

Eliminate agents with wealth = 0 (par 11)

Increment age $\text{agent}(i, 7) = \text{agent}(i, 7) + 1$

Use of resource 1 and 2 $\text{agent}(i, 11) - \text{BRC1}; \text{agent}(i, 12) - \text{BRC2}$

Assessment of GDP $\text{GDP} = \text{GDP} + \text{agent}(i, 11)$

Show and Plot

Plot Shows the agents according to their total resources (Food+Money) indicated as the sqr of the diameter.

The high of the agent is proportional to the total wealth of the agent's money.

The colour of the bubble depends on the type of agent as indicated at the left bottom of the screen.

The thickness of the border is proportional to the perceived cost of living (Price for food + Price for minerals).

The colour of the border is more reddish or even yellow the higher the ratio $\text{MinPrice}/\text{FoodPrice}$. Blue borders indicates that food prices are higher than mineral prices.

Black bar at the bottom indicates a length of 100 pixel

Internal parameters

General parameters (number in parentheses indicates the column in the master matrix)

X (0)	Spatial dimension1	
Y (1)	Spatial dimension2	
CRa (2)	Contact Radius or	Maximum distance at which interchange
	Contact Horizon	between agents may occur (e.g. altruistic interchange)
TMo (4)	Type of Movement	Type of spatial displacement: NO MOVEMENT

Characteristics of Agents

Age (7)	Age	Age of agent
WT (10)	Wealth-Money	Total capital in liquid money
WFO (11)	Wealth-Food	Amount of resource 1 (Renewable).
		If $\text{WFO}(i)=0$ then agent i starves (i is eliminated)
WCo(12)	Wealth-Commodity	Amount of resource 2
		(Minerals or Non-Renewable resource)
Dept (13)	Dept	Accumulated Dept
Well (19)	Well-being	Amount of resources $3 = r1 * r2 * \text{Gain}$
TAg (20)	Type of Agent	Specialization or task of agent
	0 Omnipotent: exploits resources 1 and 2	
	1 Farmer: exploits only resources 1	
	2 Miners: exploits only resources 2	
	3 Trader: does not collect but trades and provides credit, i.e. creates money	

External Parameters

General

Initial Nr. of Agents (ino).

Optimum Population Size (ops): Maximum number of agents aimed at through ssconst.

Simulation Scenario: production of new agents (ssconst).

0: New agents are created each time step, until the number indicated by ops is achieved. New agents are assigned internal parameters at random.

Proportion Culled (PC). The proportion of agents killed randomly when population in excess of ops.

Dangers, other than starvation danger. Large values increase random selection; large WCo reduces this.

Fitness function: Agents, in order to continue in the virtual world, had to satisfy each time step the rule: $100 * \text{Rnd}/(\text{amount of resource } 2)/(\text{mean wealth of resource } 2) < 1000 * \text{Rnd}$.

Dangers: probability of being eliminated in random selection events.

Resources 1 and 2

Resource 1 (provides WFo) and 2 (provides WCo).

Number of patches of Resource (RNR) Number of resource patches.

Size of patch of Resource (SNR) Maximum size of each patch (but see Mutation).

Degradation of Resource due to consumption (DNR). Amount lost due to consumption (RD).

Distribution Pattern of Resource (DPR) Resource is distributed:

1. Fixed size, randomly distributed
2. Fixed size, centered

Else random size, randomly distributed.

Basal rate of metabolism (BRC) Amount of resource passively used up (b).

Efficiency of consumption (EfC) Amount of resource assimilated.

When $\text{EfC} > 10$ then productivity is simulated: $\text{Agent collects} = (\text{EfC} - 10) * \text{price}$ (either for resource 1 and/or 2).

Frequency of change in distribution (FCh) Frequency in t -steps distribution changes.

Consumption of resource: Rate of exploitation: $\text{resource} = \text{resource} - \text{DNR}$.

Passive use of resource by agent (BRC): Wearing or passive use of resource agent- BRC.

Resource 1 can be modelled as a renewable resource (agriculture for example), whereas resource 2 as a nonrenewable resource (e.g. mining), by assigning $\text{DNR} = 0$ and $\text{DNR} = 1$ respectively.

Food Reserve (FR): Minimum amount of food needed for agent to engage in transactions of any kind.

Min Food for Reproduction (MFR): The amount of food that needs to be accumulated before reproduction can start when Simulation Scenario is 1 or 4.

Type of economy (EconoT)

0: No Barter nor any other interactions except taxes.

1: **Barter**: with no money.

>1: Money as Species. If **Price adjust**=0 then Fixed prices.

If **Price Adjust** > 0 the Prices are determined by demand: agents not selling decrease price by one unit; agents not finding seller from which to buy increase price by one unit.

3: +Financial: Traders lend money.

4: +Agents finishing a successful sale, increase price by one unit.

5: +Successful buyers decrease their future asking price by one unit.

6: +As in EconoT 4 and 5.

7: +As in 2, 3, 4 and 5.

8: 2 is > 1 but less than 3

10: Agents pay Taxes: Taxes collected are increased synergistically by Taxpool * SSTax prior to their distribution.

Food price: In integer units.

Mineral price: Price of commodities in integer units.

Price Adjustment: Units prices are reduced or increase when trade fails. 0 = fixed prices.

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